Jaya Sravan Kala

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years building production computer vision systems for detection, tracking, and segmentation. At Sam's Club I develop, fine-tune, and evaluate YOLO-based perception models in PyTorch running across 400+ cameras -- reaching 97% mAP with MLflow experiment tracking and versioned, regression-checked releases. I build the continuous-learning loop: automated data mining and ingestion with quality validation that lifted dataset diversity 35% while halving retraining frequency, plus drift detection that surfaces annotation issues before they reach production. M.S. in Computer Science; I ship models from training to monitored production.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, JavaScript

Machine Learning & Computer Vision: PyTorch, YOLOv11 (Detection, Segmentation, OBB), Grounding DINO, OpenCV, Hugging Face Transformers, DOLG Embeddings, scikit-learn, EasyOCR, PaddleOCR

Training, Evaluation & Optimization: MLflow (Experiment Tracking, Model Registry), Model Fine-Tuning, Evaluation Metrics (mAP, IoU), ONNX Runtime, TorchScript, Model Quantization, CI/CD Pipelines

Cloud & Data: Azure (Service Bus, Blob, Cosmos DB), Apache Kafka, AWS S3, PySpark, Delta Lake

PROFESSIONAL EXPERIENCE

Sam's Club | Computer Vision Engineer

Feb 2025 – Present

- Develop, fine-tune, and evaluate production YOLOv11 detection, segmentation, and OBB models end to end in PyTorch with MLflow experiment tracking — reaching 97% mAP on validation across models serving 400+ in-store cameras.
- Built evaluation and active-data-selection pipelines that guard production quality: an automated QA system catching annotation drift via cosine-similarity thresholds against rolling embedding templates, plus a segmentation post-processing stage (NMS, Mask IoU filtering) that cut duplicate detections by ~40%.
- Own the continuous-learning loop for perception models: automated image mining and ingestion with quality validation (blur detection, fisheye filtering) lifted dataset diversity 35% while cutting retraining frequency in half; fine-tuned models ship through the MLflow Model Registry with versioned, regression-checked releases.
- Architected a modular perception pipeline spanning 400+ shelf cameras across six deployment phases, combining YOLOv11, DOLG embeddings, VLM-based change detection, and Kafka event streaming for real-time monitoring.
- Shipped a multi-model detection pipeline (YOLOv11 + Grounding DINO, tiled inference on 3840x2160 images) with a multi-pass OCR workflow (EasyOCR to PaddleOCR to fuzzy digit normalization) for reliable extraction and severity-tiered alerting.
- Optimized models for efficient inference via quantization and ONNX Runtime / TorchScript conversion for compute-constrained deployment.

UCode Technologies LLC | Software Engineer

Sep 2024 – Jan 2025

- Built ETL pipelines ingesting structured data from SQL databases and AWS S3 into clean, model-ready datasets for predictive analytics.
- Automated data preparation workflows, cutting manual preprocessing effort and speeding up experimentation for the data science team.
- Trained and evaluated Random Forest classifiers, tuning precision-recall tradeoffs to meet business performance targets.

Cognizant | Program Analyst

Feb 2021 – Jul 2022

- Ran and monitored enterprise ETL workflows across MySQL and Snowflake, keeping data available and within SLA.
- Wrote SQL transformations and Bash automation on Delta Lake to improve data quality and shorten downstream dashboard processing time.
- Helped optimize Snowflake schemas and warehouse design, improving reporting efficiency.

PROJECTS

Multimodal Fake News Detection

- Built a fake-news detector combining CLIP and ViLT embeddings to spot text-image inconsistencies through cross-modal cosine similarity, with attention-based visualizations explaining each prediction. (PyTorch, Hugging Face Transformers)

Formula One Data Analytics Pipeline

- Built an end-to-end Azure pipeline (Data Factory, PySpark, Delta Lake) processing multi-season Formula One telemetry into Azure Data Lake Storage, with schema validation and partition tuning that sharply cut dashboard query times.

EDUCATION

George Mason University, Fairfax, VA
M.S., Computer Science — Aug 2022 to May 2024

Institute of Aeronautical Engineering, Hyderabad, India
B.Tech., Information Technology — Aug 2017 to May 2021

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate (DP-203)
- AWS Certified Cloud Practitioner